Climate and Tourism in Europe

Are Cooler Destinations Becoming More Attractive?

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Abstract

Tourism activity in Europe exhibits substantial spatial and temporal heterogeneity, partly associated with climatic conditions. This paper investigates the relationship between climate variables and tourism activity across European NUTS2 regions over the period 2010–2023 using a panel dataset combining Eurostat tourism indicators, ERA5 climate data, and regional socioeconomic variables.

The analysis primarily relies on panel econometric methods, including pooled and fixed-effects models. Supplementary machine learning and spatial network analyses are reported in the Appendix as exploratory extensions.

The results indicate that temperature anomalies have a positive and statistically significant within-region association with tourism activity, whereas precipitation does not display a robust relationship across specifications. The econometric models also suggest an inverted U-shaped relationship between temperature anomalies and tourism intensity. However, the explanatory power of climate variables remains limited relative to structural regional factors, particularly GDP.

Overall, the findings suggest that climate influences tourism dynamics at the margin, but does not fundamentally reshape regional tourism patterns once structural regional heterogeneity is accounted for. In particular, the analysis does not provide strong evidence of a generalised shift in tourism demand toward cooler European regions during the observed period.

Keywords: tourism activity, climate, temperature anomalies, panel data, fixed effects, Europe

Word count: 7,000

1 Introduction

Tourism represents a major component of economic activity across Europe, contributing to regional development, employment, and mobility. However, tourism activity remains highly uneven across space and time, with strong seasonal fluctuations and persistent regional disparities.

Climate conditions are often considered important determinants of tourism patterns. Temperature and precipitation influence destination attractiveness, travel decisions, and the feasibility of outdoor and leisure activities. Warmer and drier conditions are commonly associated with increased tourism demand in coastal and recreational regions.

Despite this apparent relationship, the empirical literature remains inconclusive regarding the magnitude and form of climate effects on tourism activity. Climate variables are strongly correlated with seasonal cycles and regional structural characteristics such as economic development, infrastructure, accessibility, and geographical attractiveness, making their independent contribution difficult to isolate.

This raises a broader question: to what extent do climate conditions structure tourism activity across European regions, and can rising temperatures generate systematic shifts in regional tourism patterns?

To address this question, this study primarily relies on panel econometric models using a harmonised NUTS2-level dataset covering the period 2010–2023. Pooled and fixed-effects specifications are used to estimate the association between climate conditions and tourism activity while accounting for regional heterogeneity and common time trends. Supplementary machine learning and spatial network analyses are reported in the Appendix as exploratory extensions rather than as the main basis for inference.

This study contributes to the literature in three ways. First, it provides a harmonised regional panel dataset combining Eurostat tourism indicators, ERA5 climate data, and socioeconomic variables across European regions. Second, it shows that climate variables—particularly temperature anomalies—are statistically associated with regional tourism activity once regional heterogeneity is taken into account, although their explanatory power remains limited relative to structural economic factors. Third, it introduces tourism market share as a robustness check to assess whether the findings depend on the use of absolute tourism volumes.

Overall, the findings suggest that climate influences tourism dynamics primarily at the margin rather than as a central determinant of regional tourism patterns. In particular, the analysis does not provide strong evidence of a generalised shift in tourism demand toward cooler European regions over the study period.

Rather than attempting to establish causal effects, the paper focuses on robust statistical associations and on the relative importance of climate variables compared with structural economic factors. All data processing and analysis steps are fully reproducible and documented in Section 9.

2 Background and Related Work

2.1 Climate Change and Tourism Dynamics

Climate change constitutes one of the most significant structural transformations affecting socio-economic systems worldwide. In Europe, rising temperatures, increasing frequency of extreme weather events, and shifting precipitation patterns are expected to alter the spatial and temporal distribution of economic activities (IPCC, 2021). As a climate-sensitive sector, tourism is particularly exposed to these changes, as environmental conditions directly influence destination attractiveness, travel decisions, and the feasibility of tourism-related activities.

Recent contributions highlight that climate change is not only a long-term environmental issue but also a key economic factor shaping regional competitiveness and sectoral dynamics. In the tourism sector, these changes may affect both the supply side, through infrastructure and environmental constraints, and the demand side, through changing preferences and travel patterns (Scott et al., 2016). As a result, understanding the relationship between climate conditions and tourism activity has become an increasingly important research topic.

2.2 Climate as a Determinant of Tourism Demand

A substantial body of research has examined the relationship between climate conditions and tourism demand. Early contributions focused on the role of weather and climate as key determinants of destination attractiveness, particularly for leisure tourism (Maddison, 2001; Lise and Tol, 2002). These studies generally find that tourists exhibit strong preferences for specific climatic conditions, typically favouring warm and dry environments.

Building on this framework, Hamilton et al. (2005) and Bigano et al. (2006) developed macro-level models of international tourism flows, showing that climate variables significantly shape global tourism patterns. These approaches emphasise the importance of climatic conditions in determining the spatial allocation of tourism demand at the global scale.

More recent work has expanded this perspective by incorporating climate change projections and exploring their long-term implications. Scott et al. (2012, 2019) argue that rising temperatures may lead to a geographical redistribution of tourism demand, potentially reducing the attractiveness of already warm regions while benefiting cooler destinations. Empirical analyses also suggest that the relationship between temperature and tourism is non-linear, with diminishing or even negative effects at high temperature levels (Rutty and Scott, 2010; Falk, 2014). These findings indicate that climate effects are complex and may vary across contexts and temperature ranges.

2.3 Climate vs Weather: Conceptual Distinction

An important conceptual distinction in the literature concerns the difference between weather and climate. Weather refers to short-term atmospheric conditions, while climate captures long-term statistical patterns. Tourism decisions may respond differently to these two dimensions. Short-term weather shocks can influence immediate travel behaviour, whereas long-term climate conditions shape destination attractiveness and structural tourism patterns (Dell et al., 2014).

From an empirical perspective, this distinction has important implications. Many studies rely on high-frequency weather data to capture short-term effects, while others focus on long-term climate averages. However, these approaches may capture different underlying mechanisms. In particular, climate variables are often correlated with geographic characteristics and persistent regional attributes, making it difficult to disentangle their independent effect from structural factors (Dell et al., 2014; Burke et al., 2015).

In this study, the use of annual aggregated climate data places the analysis closer to a climatic perspective, focusing on medium-term conditions rather than short-term weather fluctuations. This choice reflects both data availability constraints and the objective of analysing broad regional patterns across Europe.

2.4 Heterogeneity of Climate Effects

Another important dimension concerns the heterogeneity of climate effects across different types of tourism and regions. Tourism demand is not homogeneous and may respond differently to climatic conditions depending on the nature of destinations (Falk, 2014; Scott et al., 2019). Coastal regions, for instance, may benefit from higher temperatures up to a certain threshold, while urban tourism may be less sensitive to climatic variations. Similarly, mountain regions may exhibit distinct seasonal dynamics, where temperature increases can have both positive and negative effects depending on the context.

This heterogeneity implies that aggregate estimates may mask substantial variation in climate-tourism relationships. It also motivates the use of regional-level data, as adopted in this study, to better capture spatial differences in responses to climate conditions. By focusing on NUTS2 regions, the analysis allows for a more granular examination of these heterogeneous effects across Europe.

2.5 Empirical and Methodological Limitations

Despite these advances, several limitations persist in the empirical literature. First, many studies rely on aggregated national data or focus on specific destinations, limiting the ability to capture regional heterogeneity within countries (Falk, 2014). Yet tourism systems are inherently localised, and responses to climate conditions may vary substantially across regions depending on infrastructure, economic structure, and tourism specialisation.

Second, a large share of the literature relies on relatively low-frequency data (annual or seasonal aggregates), which may obscure important intra-annual dynamics (Baum and Lundtorp, 2001). Tourism seasonality is indeed a central feature of the phenomenon (Butler, 1994), and higher-frequency data would in principle allow for a more precise analysis of short-term co-movements between climate conditions and tourism activity.

However, the use of high-frequency data in a comparative regional setting raises substantial challenges. Harmonised datasets combining tourism and climate indicators at fine temporal resolution remain limited in terms of spatial coverage and cross-country comparability. As a result, large-scale comparative analyses often rely on aggregated data to ensure consistency across regions.

From a methodological perspective, the use of annual data is not only a constraint but also reflects a deliberate trade-off. By focusing on lower-frequency variation, the analysis abstracts from short-term weather fluctuations and seasonal noise, allowing for a clearer identification of medium-term relationships between climate conditions and tourism activity. This approach is particularly suited to panel data frameworks exploiting both cross-sectional and temporal variation while controlling for unobserved heterogeneity.

Observational studies also face important identification challenges. Climate variables are strongly correlated with seasonal cycles and may co-evolve with unobserved factors influencing tourism demand, such as economic conditions, infrastructure development, or policy interventions (Falk, 2014). In addition, tourism development itself may affect regional characteristics including accessibility, infrastructure, and economic activity, raising potential endogeneity concerns.

These issues complicate causal interpretation, as climate conditions remain intertwined with broader geographic, economic, and institutional factors (Angrist and Pischke, 2009). In this context, annual aggregates may help mitigate part of the short-term variability and seasonal noise, although they do not fully resolve identification concerns. As a result, recent contributions emphasise the importance of transparent empirical strategies and careful interpretation when relying on non-experimental data (Salganik, 2017).

2.6 Research Gap and Contribution

Taken together, these limitations point to a specific empirical gap in the literature. While existing studies have established that climate conditions are associated with tourism activity, the joint analysis of tourism, socioeconomic, and climate variables at a comparable regional scale remains constrained by data availability and cross-country comparability. In particular, harmonised datasets combining tourism indicators, regional economic variables, and climate information at the NUTS2 level are relatively scarce, limiting the scope for large-scale comparative analysis across Europe.

A second limitation concerns the distinction between cross-sectional differences and within-region variation over time. Many observed relationships between climate and tourism may reflect persistent regional characteristics, such as geography, infrastructure, accessibility, or economic development, rather than the effect of climate fluctuations themselves. This creates a need for empirical frameworks that can separate structural regional differences from temporal variation while remaining transparent about the limits of causal interpretation.

Given the observational nature of the available data, identifying causal effects is inherently challenging. Climate variables co-vary with seasonal patterns, regional attractiveness, economic conditions, infrastructure, and other determinants of tourism demand. As a result, this study does not aim to estimate causal effects, but rather to document robust statistical associations between climate conditions and regional tourism activity within a consistent panel framework.

The present study addresses these gaps by constructing a harmonised panel dataset at the NUTS2 level covering European regions over the period 2010–2023. By combining tourism indicators from Eurostat with ERA5 climate data and regional socioeconomic variables, the analysis enables a systematic comparison of climate-tourism relationships across European regions within a unified empirical framework.

Methodologically, the paper primarily adopts a panel econometric approach. Pooled and fixed-effects specifications are used to exploit both cross-sectional and temporal variation while controlling for unobserved regional heterogeneity and common time trends. This design makes it possible to distinguish broad structural differences between regions from within-region associations between climate fluctuations and tourism activity.

The paper contributes to the literature in three main ways. First, it provides a reproducible harmonised regional dataset linking tourism, socioeconomic, and climate information across Europe. Second, it offers panel-based evidence on the relationship between temperature, precipitation, and tourism activity while explicitly accounting for regional heterogeneity. Third, it introduces tourism market share as a robustness check to assess whether the findings depend on the use of absolute tourism volumes rather than relative regional tourism performance.

Supplementary machine learning and spatial network analyses are reported in the Appendix. These extensions provide additional descriptive insights into predictive patterns, non-linearities, and territorial structure, but they are not the primary basis for inference. The central contribution of the paper rests on the panel econometric analysis of climate-tourism relationships across European regions.

3 Research Problem

This study addresses the following research question:

> What is the relationship between climate conditions and tourism activity across European regions over time?

The analysis adopts a relational rather than causal perspective, focusing on the identification of statistical associations between climate variables and tourism activity. Given the observational nature of the data and the absence of a clear identification strategy, the objective is not to estimate causal effects but to provide a robust empirical characterisation of climate-tourism relationships.

To address this question, the study adopts a longitudinal panel design exploiting both cross-sectional variation across regions and temporal variation over the period 2010–2023. Although the use of annual data limits the analysis of intra-annual dynamics, it ensures consistency and comparability across regions while allowing the identification of medium-term tourism patterns.

Identification Challenges

Several identification challenges arise when analysing the relationship between climate conditions and tourism activity. First, omitted variable bias may occur if factors such as infrastructure, accessibility, or tourism specialisation are not fully captured in the model. These variables are likely correlated with both climate conditions and tourism demand, potentially biasing estimated relationships.

Second, reverse causality and simultaneity cannot be entirely ruled out. While climate conditions may influence tourism activity, tourism development may itself affect regional characteristics such as infrastructure and economic activity.

Third, climate variables are strongly correlated with seasonal and broader time-varying dynamics, making it difficult to isolate their independent contribution. More generally, these issues reflect the broader identification problem in observational data, where explanatory variables co-vary with unobserved determinants of the outcome (Angrist and Pischke, 2009).

As a result, the analysis focuses on identifying robust statistical relationships rather than causal effects, in line with recent approaches in computational social science (Salganik, 2017).

Based on the existing literature, two empirical expectations guide the analysis:

H1: Tourism activity is positively associated with higher temperatures and negatively associated with precipitation.

H2: The strength of this relationship varies across regions, reflecting differences in tourism specialisation, environmental conditions, and regional characteristics.

These hypotheses are formulated in relational terms and are tested within a panel data framework controlling for unobserved regional heterogeneity and common temporal trends.

4 Data

The data and code used in this study are publicly available in a GitHub repository, ensuring full reproducibility of the analysis. The study combines multiple data sources to construct a harmonised regional panel dataset integrating economic, demographic, environmental, and geographic information at the NUTS2 level.

Tourism data are obtained from Eurostat and measure the annual number of nights spent in accommodation establishments. After filtering for total nights, accommodation services, and NUTS2 regions, the resulting dataset contains 4,365 region-year observations.

Gross domestic product (GDP) data are retrieved from Eurostat and filtered to retain total GDP expressed in million euros. Population data are also obtained from Eurostat and filtered for total population. After harmonisation and reshaping, these datasets contain 7,498 and 9,509 observations respectively.

Climate data are derived from the ERA5 reanalysis dataset provided by the Copernicus Climate Data Store. Temperature and precipitation variables are aggregated to annual values and spatially matched to NUTS2 regions using geographic coordinates and regional boundaries. Temperature is measured as an annual average, while precipitation is aggregated as total annual accumulation.

The use of annual data reflects both methodological and data availability constraints. Although higher-frequency data would allow a more detailed analysis of seasonal dynamics, harmonised tourism and climate information at the NUTS2 level remains limited in terms of temporal coverage and cross-country comparability. Annual aggregation therefore provides a consistent framework for comparative regional analysis while remaining aligned with the study's focus on medium-term climate conditions.

The NUTS2 geographic boundaries used for spatial aggregation are obtained from the GISCO database provided by Eurostat.

The analysis covers the period from 2010 to 2023. This time window was selected because it represents the longest interval for which tourism, economic, demographic, and climate information could be consistently matched across European NUTS2 regions while maintaining a stable regional classification framework. Although some individual datasets contain observations before 2010, combining all variables substantially reduces geographical coverage and consistency in earlier years. Starting the analysis in 2010 therefore maximises regional coverage while preserving comparability across regions and data sources. The selected period also spans several major European heatwave episodes and provides sufficient temporal variation for empirical analysis.

The integration of these heterogeneous sources requires the alignment of regional identifiers, temporal coverage, and spatial resolution. The intersection of the economic and demographic datasets yields 293 regions prior to the inclusion of climate information.

After merging all data sources and excluding observations with missing climate values, the final dataset covers 273 European regions across 34 countries over the period 2010–2023, resulting in 3,274 observations.

The resulting dataset forms an unbalanced panel with a coverage ratio of approximately 85.7% of all possible region-year observations. By combining tourism, economic, demographic, environmental, and geographic information within a consistent regional framework, the dataset provides a comprehensive basis for analysing the relationship between climate conditions and regional tourism activity across Europe.

5 Operationalisation

The dependent variable is tourism activity, measured as the annual number of nights spent in accommodation establishments at the regional level. This indicator is widely used in the tourism economics literature as a proxy for regional tourism activity and destination attractiveness. Given the strong right-skewness of this variable, a logarithmic transformation is applied to reduce heteroskedasticity and facilitate interpretation in relative terms.

The main explanatory variables capture climatic conditions. Temperature is measured as the annual average (in degrees Celsius), while precipitation corresponds to total annual accumulation (in millimetres). Both variables are treated as continuous predictors reflecting the overall climatic environment within each region-year.

Additional control variables include gross domestic product (GDP) and population, capturing economic development and demographic scale. These controls account for structural differences in tourism demand and regional capacity.

All variables are defined at the NUTS2 regional level and measured on an annual basis. The resulting dataset combines cross-sectional variation across regions with temporal variation over the period 2010–2023, providing a suitable framework for analysing the relationship between climate conditions and regional tourism activity across Europe.

For the econometric models, the logarithmic specification implies that estimated coefficients can be interpreted as semi-elasticities, relating changes in climate variables to percentage variations in tourism activity. In addition, quadratic temperature terms are introduced in selected specifications to investigate potential non-linear climate-tourism relationships suggested by the literature.

5.1 Limitations

The tourism variable used in this study measures aggregate regional tourism activity through annual overnight stays. While widely employed in the tourism economics literature, this indicator does not capture tourist origins, travel motivations, expenditures, or inter-regional mobility flows.

Consequently, the results should be interpreted as changes in regional tourism activity rather than direct evidence of tourist relocation across Europe. The analysis therefore focuses on the relationship between climate conditions and tourism intensity at the regional level. Future research could complement this approach with origin–destination data or more disaggregated tourism indicators to better investigate mobility patterns and spatial redistribution effects.

6 Analytical Approach

The empirical strategy relies primarily on panel data econometric techniques to analyse the relationship between climate conditions and regional tourism activity across European regions. The objective is to exploit both the cross-sectional variation between regions and the temporal variation observed over the period 2010–2023 while accounting for unobserved heterogeneity and common time trends.

6.1 Baseline and Panel Specifications

The baseline specification is estimated using ordinary least squares with time fixed effects:

$$\log(Tourism_{r,t}) = \beta_1 Temperature_{r,t} + \beta_2 Precipitation_{r,t} + \beta_3 \log(GDP_{r,t}) + \beta_4 \log(Population_{r,t}) + \lambda_t + \epsilon_{r,t}$$

This specification captures both cross-sectional and temporal variation while controlling for common time shocks affecting all regions.

To account for unobserved heterogeneity across regions, a fixed effects panel model is estimated:

$$\log(Tourism_{r,t}) = \beta_1 Temperature_{r,t} + \beta_2 Precipitation_{r,t} + \alpha_r + \lambda_t + \epsilon_{r,t}$$

where α_r captures region-specific effects and λ_t common time effects. Standard errors are clustered at the regional level to account for within-region correlation over time.

This specification exploits within-region variation over time, allowing the identification of the association between climate fluctuations and tourism activity relative to region-specific averages.

6.2 Non-linear Specification

To account for potential non-linear effects of temperature, an alternative specification includes a quadratic term:

$$\log(Tourism_{r,t}) = \beta_1 Temp^*_{r,t} + \beta_2 (Temp^*_{r,t})^2 + \beta_3 Precipitation_{r,t} + \alpha_r + \lambda_t + \epsilon_{r,t}$$

where $Temp^*_{r,t}$ denotes temperature anomalies, defined as deviations from the regional mean.

This transformation removes cross-sectional differences in climate levels and focuses on deviations from typical local conditions. The quadratic specification allows identifying potential threshold effects in the relationship between temperature and tourism activity.

The implied optimal temperature is derived as:

$$T^* = -\frac{\beta_1}{2\beta_2}$$

6.3 Relative Tourism Activity

To further assess the robustness of the results, an additional indicator of relative tourism activity is constructed.

For each region-year observation, tourism market share is defined as:

$$Share_{r,t} = \frac{Tourism_{r,t}}{\sum_j Tourism_{j,t}}$$

where the denominator corresponds to total tourism activity observed across all European regions in year t.

Although this measure does not capture individual mobility flows directly, it provides a relative measure of regional tourism performance within the European tourism system. Additional fixed-effects specifications are therefore estimated using the logarithm of tourism market share as the dependent variable.

Because tourism market share differs from tourism volume only by a year-specific scaling factor, the market share specifications are interpreted primarily as robustness checks designed to assess whether the main findings depend on the use of absolute or relative tourism measures.

6.4 Interpretation Framework

Given the observational nature of the data, the analysis focuses on identifying statistical associations rather than establishing causal relationships. The econometric framework is designed to account for regional heterogeneity and common temporal shocks while investigating both linear and non-linear relationships between climate conditions and regional tourism activity.

Tourism activity is measured through annual overnight stays in accommodation establishments. Consequently, the results should be interpreted as relationships between climate conditions and regional tourism intensity rather than direct evidence of tourist mobility or relocation across Europe. Additional specifications based on tourism market share are used as robustness checks to assess whether the findings depend on the use of absolute or relative tourism measures.

Supplementary machine learning and network analyses are reported in the Appendix as exploratory robustness and descriptive exercises. These complementary approaches provide additional insights into data structure and predictive patterns but do not constitute the primary basis for inference.

The main contribution of the study therefore rests on the panel econometric analysis of climate-tourism relationships across European regions.

7 Results

7.1 Descriptive Patterns

Figure 1 presents the distribution of tourism activity across regions in logarithmic scale. The transformation produces a relatively symmetric distribution, indicating that the log specification effectively reduces skewness and stabilises variance.

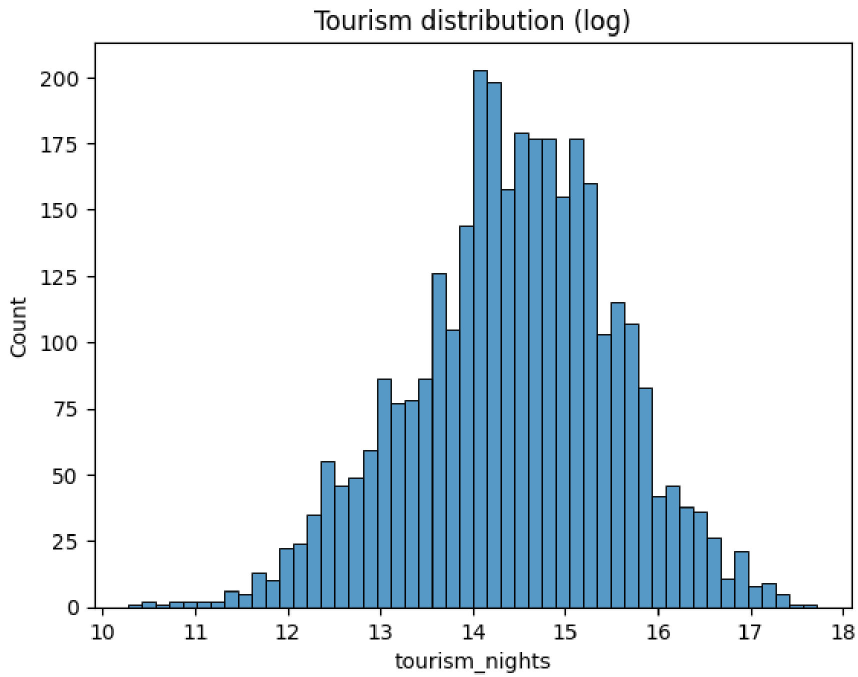

Figure 1: Distribution of tourism activity (log scale)

Figures 2 and 3 display the spatial distribution of temperature and tourism activity across European regions. A clear north–south gradient is observed for temperature, with warmer conditions in Southern Europe. Tourism activity is also relatively high in several southern and coastal regions, suggesting a positive spatial association between favourable climatic conditions and tourism intensity.

However, this relationship is not uniform across space. Some regions with comparable climatic conditions exhibit substantially different levels of tourism activity, indicating that climate alone does not explain regional tourism patterns. This points to the role of additional structural factors, such as economic development, infrastructure, accessibility, cultural attractiveness, and tourism specialisation.

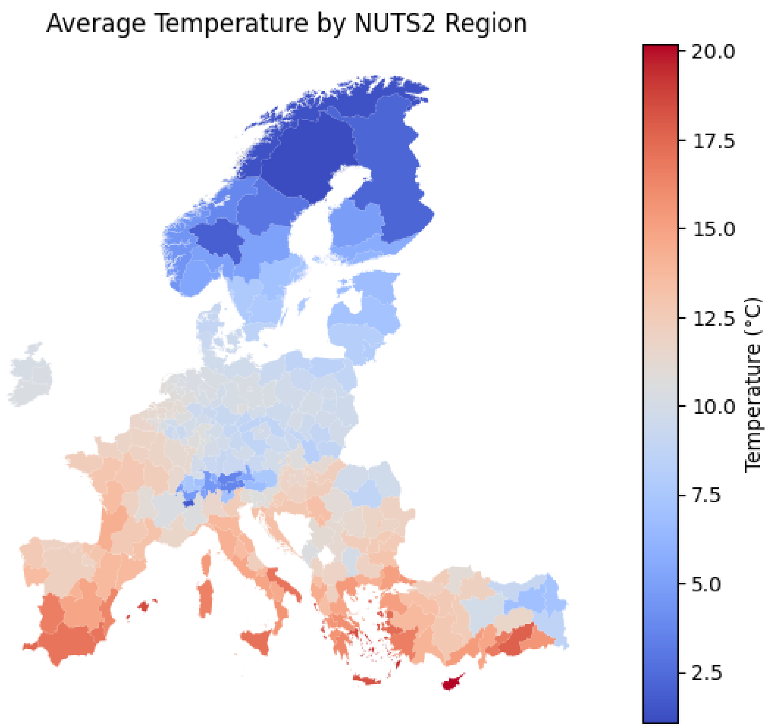

Figure 2: Average temperature by NUTS2 region

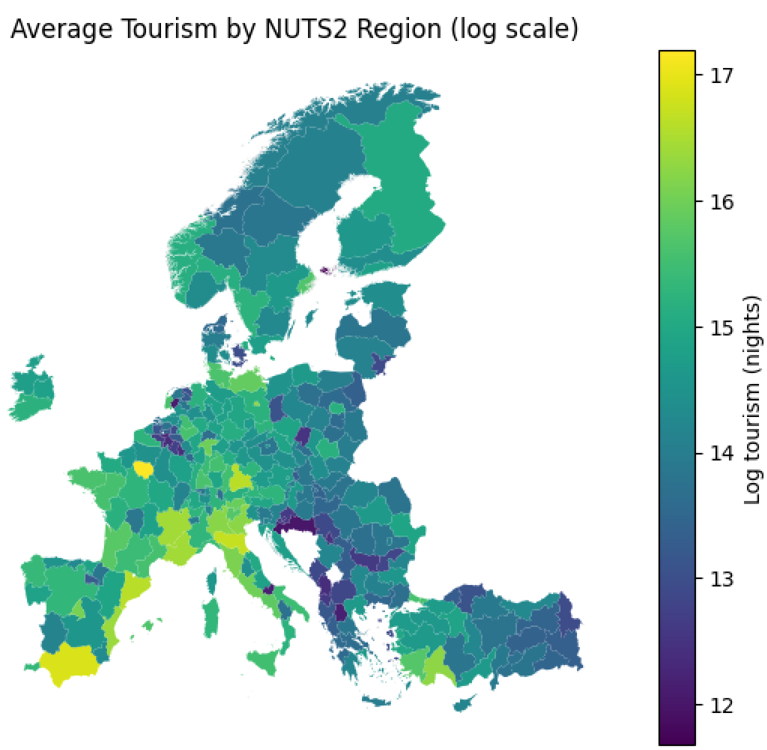

Figure 3: Average tourism activity by NUTS2 region (log scale)

Figure 4 shows the evolution of average temperature over time. A clear upward trend is observed over the period 2010–2023, with an increase of approximately 2–2.5°C across the sample.

This magnitude should be interpreted cautiously. It reflects aggregated regional averages within the constructed sample and should not be read as a direct estimate of long-run climate change. Differences in spatial coverage, aggregation methods, and panel composition may contribute to the observed variation.

Nevertheless, the figure indicates that climatic conditions changed substantially over the study period. This supports the relevance of analysing the relationship between temperature variation and regional tourism activity.

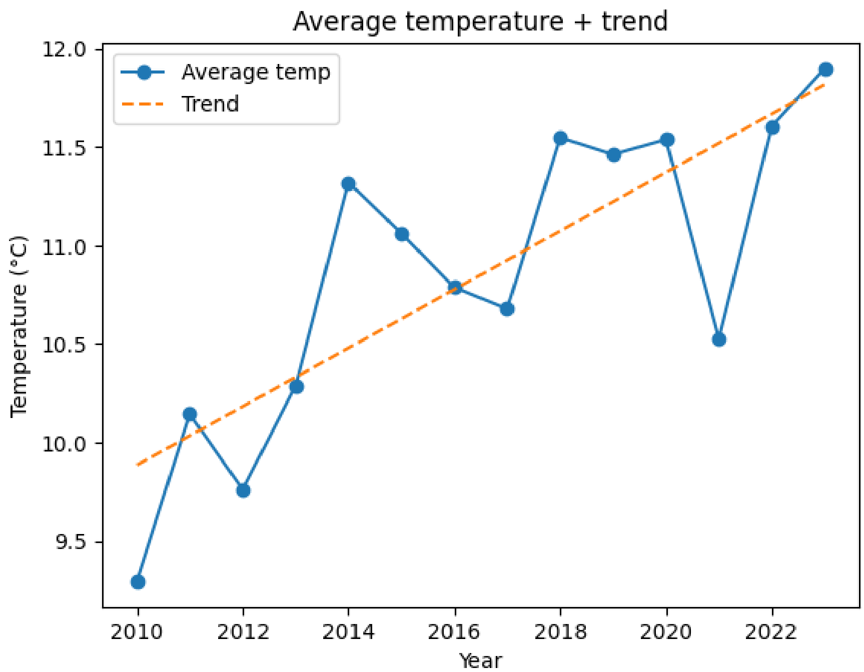

Figure 4: Average temperature over time

7.2 Baseline and Panel Estimates

Table 1 reports the results from the OLS and fixed-effects panel regressions. The specifications differ in the type of variation they exploit, allowing for a comparison between cross-sectional patterns and within-region dynamics over time. In the OLS model, GDP and population are included as control variables capturing structural differences in economic size and tourism capacity across regions.

The OLS results indicate that economic variables are the main drivers of tourism activity. GDP exhibits a strong and statistically significant association with tourism activity ($\beta = 0.620$, $p < 0.01$), while population also has a positive effect ($\beta = 0.174$, $p < 0.1$). The coefficient on temperature is positive but relatively small and only marginally significant ($\beta = 0.034$, $p \approx 0.05$). Precipitation appears positive and statistically significant in this cross-sectional specification ($\beta = 6.13 \times 10^{-5}$, $p < 0.01$).

Taken together, these results primarily reflect differences across regions rather than changes within regions over time. In particular, richer and more populated regions tend to exhibit higher tourism activity, and these structural differences dominate the cross-sectional relationships observed in the data.

In the fixed-effects framework, region-specific characteristics are absorbed by entity fixed effects. This means that the estimation focuses on variation within each region over time, while controlling for all time-invariant characteristics such as geography, baseline climate, regional identity, and long-term economic structure. As a result, slowly evolving variables such as GDP and population provide limited additional within-region information and are excluded from the fixed-effects specifications.

To better isolate short-term climate variation, temperature is expressed as an anomaly, defined as the deviation from the region-specific mean:

$$Temp^*_{r,t} = Temp_{r,t} - \bar{Temp}_r$$

This transformation removes persistent cross-sectional differences in climate levels and allows the analysis to focus on deviations from typical local climatic conditions.

Once region fixed effects are introduced, the interpretation of climate variables changes substantially. Temperature anomalies become larger and highly significant ($\beta = 0.171$, $p < 0.01$). This suggests that, within a given region, warmer-than-usual years are associated with higher tourism activity. In semi-elasticity terms, a one-degree increase relative to usual local climatic conditions is associated with an increase of roughly 17–19% in tourism activity.

This effect should not be interpreted as evidence of long-run tourist relocation. Rather, it indicates that short-term deviations from usual regional temperature conditions are associated with changes in tourism intensity within the same region.

In contrast, precipitation loses statistical significance ($\beta = 1.50 \times 10^{-5}$, $p > 0.5$), suggesting that its apparent effect in the OLS specification is largely driven by persistent differences across regions rather than meaningful within-region variation over time.

However, the relationship between temperature and tourism is unlikely to be purely linear. To account for potential non-linear effects, a quadratic term is introduced. The results reveal a statistically significant inverted U-shaped relationship between temperature anomalies and tourism activity. The linear term remains positive ($\beta = 0.133$, $p < 0.01$), while the squared term is negative and significant ($\beta = -0.119$, $p < 0.01$), indicating diminishing returns to positive temperature deviations.

This implies that tourism activity increases with temperature anomalies up to a certain point, after which additional increases are associated with a decline in tourism activity. The estimated turning point corresponds to a temperature anomaly of approximately $+0.56°\mathrm{C}$ relative to the regional mean.

Formally, this turning point is obtained as:

$$T^* = -\frac{\beta_1}{2\beta_2}$$

where β_1 and β_2 denote the coefficients on the linear and squared temperature anomaly terms. This value should be interpreted as a statistical optimum within the observed data rather than as a universal behavioural threshold.

Despite these statistically significant effects, the overall explanatory power of climate variables remains limited. The within R^2 of the fixed-effects models is relatively low, indicating that climate fluctuations explain only a small share of intra-regional variation in tourism over time.

This result highlights the importance of structural and economic factors in shaping tourism patterns. Climate conditions matter, but their influence remains secondary compared to persistent regional characteristics and economic capacity. In particular, the magnitude of the temperature effect remains modest relative to the cross-sectional variation captured by GDP, reinforcing the conclusion that economic development and regional attractiveness are central determinants of tourism activity in the European context.

Table 1: Climate and tourism: panel regression results

Variable	(1) OLS	(2) FE	(3) FE Non-linear
Temperature	0.034*		
	(0.018)		
Temperature anomaly		0.171***	0.133***
		(0.040)	(0.032)
Temperature anomaly^2			-0.119***
			(0.028)
Precipitation	6.13e-05***	1.50e-05	-1.24e-06
	(1.82e-05)	(2.27e-05)	(2.51e-05)
log(GDP)	0.620***		
	(0.058)		
log(Population)	0.174*		
	(0.086)		
Intercept	4.563***	14.412***	14.480***
	(0.863)	(0.025)	(0.031)
Observations	3274	3274	3274
Regions	273	272	272
Year FE	Yes	Yes	Yes
Region FE	No	Yes	Yes
R^2	0.528		
Within R^2		0.133	0.169

Standard errors in parentheses, clustered at the regional level. *** p<0.01, ** p<0.05, * p<0.1. Temperature anomaly is defined as deviation from the regional mean. Year fixed effects included.

7.3 Non-linear Climate Effects

Figure 5 presents the estimated non-linear relationship between temperature anomalies and tourism activity. The results reveal a clear inverted U-shaped relationship: tourism increases with positive temperature deviations up to a threshold, after which further warming is associated with lower tourism activity.

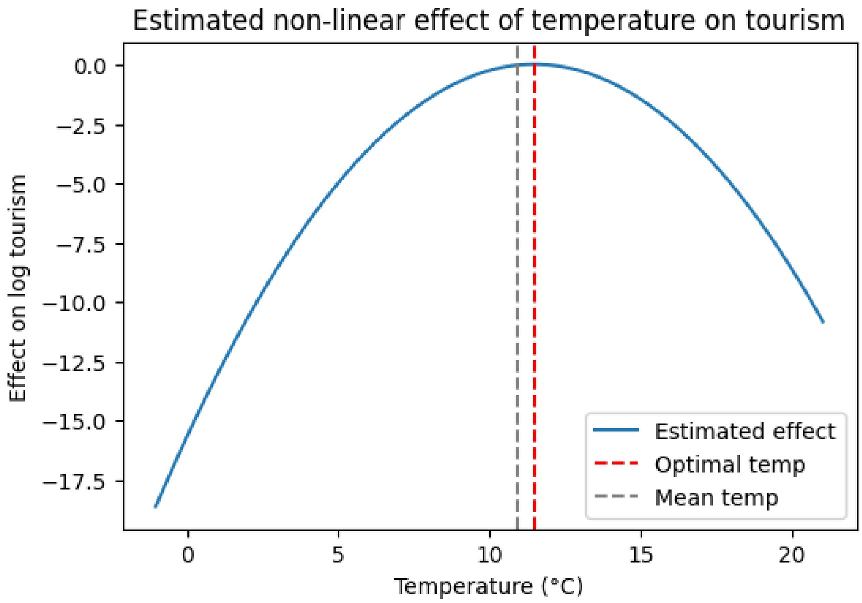

Figure 5: Estimated non-linear effect of temperature anomalies on tourism

The estimated turning point is approximately +0.56°C relative to the regional mean. This value should be interpreted cautiously, as it reflects a statistical optimum within the panel rather than an absolute preferred temperature level.

7.4 Robustness: Relative Tourism Activity

To assess whether the main findings depend on the use of absolute tourism volumes, the fixed-effects specifications are replicated using tourism market share as the dependent variable. This indicator measures each region's share of total European tourism activity in a given year.

The results remain virtually unchanged once year fixed effects are included. This is expected because tourism market share differs from tourism volume only by a year-specific scaling factor:

$$\log(Share_{r,t}) = \log(Tourism_{r,t}) - \log\left(\sum_j Tourism_{j,t}\right)$$

The second term is common to all regions within year t and is therefore absorbed by the year fixed effects. Consequently, the market share specification confirms that the main estimates do not depend on whether tourism activity is measured in absolute or relative terms.

This robustness check is particularly relevant given the aggregate nature of the tourism variable. While tourism market share does not measure individual mobility flows directly, it provides a useful alternative formulation of regional tourism performance within the European tourism system.

7.5 Synthesis of Findings

Overall, the econometric results indicate that climate conditions are associated with regional tourism activity across European regions, although their explanatory power remains limited relative to structural economic factors.

The comparison between OLS and fixed-effects specifications reveals the importance of distinguishing cross-sectional differences from within-region variation over time. In the cross-sectional framework, GDP emerges as the dominant determinant of tourism activity, while climate variables exhibit relatively modest effects. Once regional fixed effects are introduced, the analysis focuses exclusively on deviations from each region's long-run characteristics, providing a more appropriate framework for assessing the influence of climate fluctuations.

The fixed-effects estimates show that positive temperature anomalies are associated with significant increases in tourism activity. In contrast, precipitation does not exhibit a robust effect once regional heterogeneity is taken into account. The non-linear specification further suggests the existence of diminishing returns to temperature, with an inverted U-shaped relationship between temperature anomalies and tourism activity.

The robustness analysis based on tourism market share confirms that the main findings are not driven by the use of absolute tourism volumes. However, given the observational nature of the data and the aggregate measure of tourism employed, these findings should be interpreted as statistical associations between climate conditions and regional tourism intensity rather than direct evidence of tourist mobility or relocation across Europe.

Supplementary machine learning and network analyses are reported in the Appendix. These exploratory exercises provide additional descriptive insights into predictive patterns and spatial organisation, but they do not constitute the primary basis for inference. The central contribution of the study rests on the panel econometric evidence.

Taken together, the results suggest that climate influences tourism primarily at the margin. Warmer-than-usual conditions are associated with higher regional tourism activity up to a threshold, beyond which additional warming appears less favourable. Nevertheless, persistent regional differences, economic development, and structural attractiveness remain the main drivers of tourism performance across Europe.

8 Limitations

This study is subject to several limitations related to identification, data construction, measurement, and model specification.

Identification and omitted variables First, the use of observational data prevents causal interpretation. The estimated relationships should therefore be interpreted as statistical associations rather than causal effects. Although the fixed-effects framework controls for time-invariant regional characteristics and common time shocks, the results may still be influenced by omitted time-varying factors such as tourism policy, infrastructure investment, transport accessibility, accommodation capacity, or regional tourism specialisation.

Second, the analysis relies on a limited set of control variables. GDP and population capture important structural dimensions related to economic scale and demographic size, but they do not fully account for the broader determinants of tourism activity. Factors such as cultural attractiveness, accessibility, local amenities, institutional quality, or the composition of tourism demand are not directly observed in the dataset.

Temporal and spatial aggregation Third, the analysis is conducted at an annual level. This implies a loss of information on seasonal dynamics, which are central to tourism behaviour. Tourism demand is often concentrated in specific months, while temperature and precipitation effects may vary substantially across seasons. Annual aggregation may therefore obscure short-term effects that are relevant for travel decisions. This choice reflects a trade-off between temporal precision and cross-regional comparability, allowing the construction of a consistent European NUTS2 panel over the period 2010–2023.

Fourth, climate data are spatially aggregated from gridded observations to NUTS2 regions. This reduces spatial precision and may mask local heterogeneity, particularly in geographically diverse regions such as mountainous, coastal, or mixed urban-rural areas. The climate variables should therefore be interpreted as regional averages rather than precise local conditions experienced by tourists.

Measurement and interpretation Fifth, the dependent variable measures aggregate regional tourism activity through annual nights spent in accommodation establishments. While this indicator is widely used in tourism economics, it does not capture tourist origins, travel motivations, expenditure, activities, duration preferences, or inter-regional mobility flows. The analysis therefore does not directly observe whether tourists move from one region to another in response to climate conditions.

Sixth, tourism nights measure realised tourism activity rather than pure tourism demand. They may be affected by supply-side constraints such as accommodation capacity, local infrastructure, prices, and regulatory conditions. As a result, changes in overnight stays may reflect both demand-side responses and constraints on regional tourism capacity.

Seventh, tourism market share is used as a robustness check to compare absolute and relative measures of tourism activity. However, this indicator does not solve the absence of origin-destination mobility data. It provides a relative measure of regional tourism performance within the European system, but it does not identify actual tourist flows between regions.

Eighth, temperature anomalies are defined as deviations from region-specific average conditions. This improves the interpretation of within-region climatic variation, but it also means that the estimated coefficients should not be interpreted as responses to absolute temperature levels or as universal behavioural thresholds.

Model specification Finally, the results may depend on modelling choices. The econometric analysis suggests a non-linear relationship between temperature anomalies and tourism activity, with positive effects up to a threshold and declining effects beyond it. This pattern should be interpreted as a statistical relationship within the observed sample rather than as a general behavioural law.

Supplementary machine learning and network analyses are reported in the Appendix. These analyses provide exploratory insights into predictive patterns and spatial organisation, but they are not used as the primary basis for inference. In particular, the network analysis relies on geographic adjacency between NUTS2 regions and does not represent actual tourist mobility, transport flows, economic linkages, or origin-destination movements.

Taken together, these limitations imply that the findings should be interpreted as evidence of associations between climate conditions and regional tourism intensity, not as precise estimates of causal behavioural responses or direct evidence of tourist relocation. The central contribution of the study rests on the panel econometric analysis, while the supplementary analyses provide additional descriptive context.

9 Discussion and Conclusion

Summary of findings This paper analyses the relationship between climate conditions and regional tourism activity across Europe. The analysis is based on a harmonised panel dataset covering 273 NUTS2 regions across 34 European countries over the period 2010–2023, combining tourism, economic, demographic, and climate data within a common empirical framework.

The results show that climate variables are statistically associated with tourism activity, but that their explanatory power remains limited once regional heterogeneity is taken into account. In the baseline OLS specification, structural economic variables dominate the explanation of tourism activity. GDP exhibits a strong positive association with tourism, indicating that economic scale and regional capacity are central determinants of tourism intensity across European regions.

The fixed-effects specifications provide a more precise interpretation of within-region variation over time. Once time-invariant regional characteristics are controlled for, temperature anomalies become positive and statistically significant. A one-degree increase relative to usual regional conditions is associated with an increase of approximately 17% in tourism activity. In contrast, precipitation does not exhibit a robust effect once regional heterogeneity is accounted for.

The analysis also identifies a non-linear relationship between temperature anomalies and tourism activity. The econometric results suggest an inverted U-shaped pattern: warmer-than-usual conditions are associated with higher tourism activity up to a threshold, after which additional warming becomes less favourable. The estimated turning point corresponds to a temperature anomaly of approximately $+0.56°C$ relative to the regional mean. This value should be interpreted as a statistical feature of the observed panel rather than as a universal behavioural threshold.

As a robustness check, the analysis was replicated using tourism market share as an alternative dependent variable. The results remain unchanged once year fixed effects are included. This is expected because the logarithm of tourism market share differs from the logarithm of tourism volume only by a year-specific scaling factor, which is absorbed by the time effects. This confirms that the main findings do not depend on whether tourism activity is measured in absolute or relative terms.

Interpretation Taken together, the findings suggest that climate conditions influence tourism activity at the margin rather than acting as the primary determinant of regional tourism patterns. Short-term temperature deviations within regions are associated with changes in tourism intensity, but persistent differences in economic development, infrastructure, accessibility, and regional attractiveness remain central to explaining tourism performance across Europe.

The results should therefore be interpreted cautiously. The dependent variable measures annual overnight stays in accommodation establishments. This is a standard indicator of tourism activity, but it does not directly capture tourist origins, motivations, expenditure, activities, or inter-regional mobility flows. Consequently, the analysis does not provide direct evidence that tourists relocate from one region to another in response to climate conditions. It shows instead that regional tourism activity is statistically associated with climatic variation.

The non-linear specification suggests that temperature effects are not uniform across the range of observed conditions. Moderate positive temperature anomalies appear favourable to tourism activity, while excessive deviations may reduce this benefit. This finding is consistent with the idea that climate can affect destination attractiveness, but the magnitude of the effect remains limited relative to structural regional factors.

Relation to the research question In relation to the broader question motivating the study, the results provide only partial support for the idea that climate change may alter the spatial distribution of tourism in Europe. The evidence does not show a clear or systematic redistribution of tourism toward cooler regions over the period considered. Instead, the results suggest that climate-related effects are likely to be gradual, region-specific, and mediated by existing economic and territorial structures.

The tourism market share robustness check reinforces this interpretation. Since the market share specification produces equivalent estimates under year fixed effects, the main findings are not driven by the distinction between absolute tourism volumes and relative regional tourism performance. However, this robustness check does not solve the absence of origin-destination mobility data. It should therefore be understood as an alternative formulation of regional tourism intensity, not as a direct measure of tourism flows.

Methodological contribution From a methodological perspective, the study illustrates the value of panel econometric methods for separating cross-sectional regional differences from within-region dynamics over time. The comparison between OLS and fixed-effects models is particularly informative: while cross-sectional variation is dominated by economic structure, within-region variation reveals a clearer association between temperature anomalies and tourism activity.

Supplementary machine learning and network analyses are reported in the Appendix. These analyses provide additional exploratory insights into predictive patterns and spatial organisation, but they are not used as the primary basis for inference. Machine learning models are useful for examining predictive structure and non-linearities, while the network analysis provides a descriptive representation of territorial adjacency. However, neither approach directly identifies causal effects or observed tourism flows. The central contribution of the paper therefore rests on the panel econometric evidence.

Implications for climate change and tourism The results suggest that climate change may affect tourism activity, but its effects are likely to interact with existing regional structures rather than independently reshape tourism geography in the short run. Regions with strong economic capacity, tourism infrastructure, accessibility, and established attractiveness may remain competitive despite changing climatic conditions. Conversely, regions with potentially more favourable future climates may not automatically benefit if they lack the infrastructure or economic conditions required to absorb additional tourism demand.

This has important implications for regional tourism policy. Adaptation strategies should not focus only on climatic advantages or disadvantages. They should also consider infrastructure, diversification of tourism activities, seasonal management, accommodation capacity, and regional development. Climate may influence tourism margins, but regional competitiveness remains strongly embedded in broader economic and territorial conditions.

Future research Future research could extend this work in several directions. First, higher-frequency tourism and climate data would make it possible to analyse seasonal and monthly dynamics, which are central to tourism behaviour but cannot be fully captured with annual data.

Second, richer tourism indicators would improve interpretation, especially if they included tourist origins, length of stay, expenditure, activity type, or domestic versus international tourism. Third, origin-destination data would allow a direct analysis of mobility flows and would provide a stronger empirical basis for studying potential tourism redistribution across Europe.

Future research could also incorporate additional regional controls, such as transport accessibility, accommodation capacity, tourism infrastructure, local amenities, and policy interventions. Finally, quasi-experimental designs or more disaggregated spatial data could help strengthen causal identification and distinguish more clearly between climate effects, economic structure, and regional tourism capacity.

Overall, this study contributes to a better understanding of climate-tourism relationships in Europe by showing both their existence and their limits. Temperature anomalies are associated with regional tourism activity, and the relationship appears non-linear. However, the magnitude and explanatory power of climate effects remain limited relative to structural economic and regional factors. The findings therefore suggest that climate matters for tourism, but primarily as one component of a broader system of regional attractiveness rather than as a dominant force reshaping European tourism patterns.

Data and Code Availability

All data, code, and presentation materials are publicly available in the following repository:

https://github.com/lolipop913/CSS_Tourism.git

The repository includes the cleaned dataset, data processing scripts, replication code, and presentation materials used in this study.

10 Supplementary Analyses

The analyses reported in this Appendix complement the main panel econometric analysis. They are exploratory and descriptive in nature, and are not used as the primary basis for inference. Their purpose is to provide additional evidence on predictive patterns, non-linearities, and territorial structures observed in the data.

10.1 Machine Learning Evidence

While the econometric models provide structured estimates of within-region relationships, their explanatory power remains limited and they may not fully capture complex non-linear patterns. Machine learning methods are therefore used as exploratory tools to provide a flexible, data-driven characterisation of climate-tourism relationships and to detect potential threshold or extreme effects. Tree-based methods are preferred due to their interpretability and ease of implementation.

Figure 6 presents the segmentation obtained from the decision tree model. The structure is primarily driven by GDP thresholds, with the first split occurring at relatively high income levels. Regions with high GDP exhibit substantially higher tourism activity, while lower-income regions are further segmented into finer groups. Climate variables do not appear among the main splitting criteria, indicating that economic factors dominate the hierarchical structure of tourism patterns.

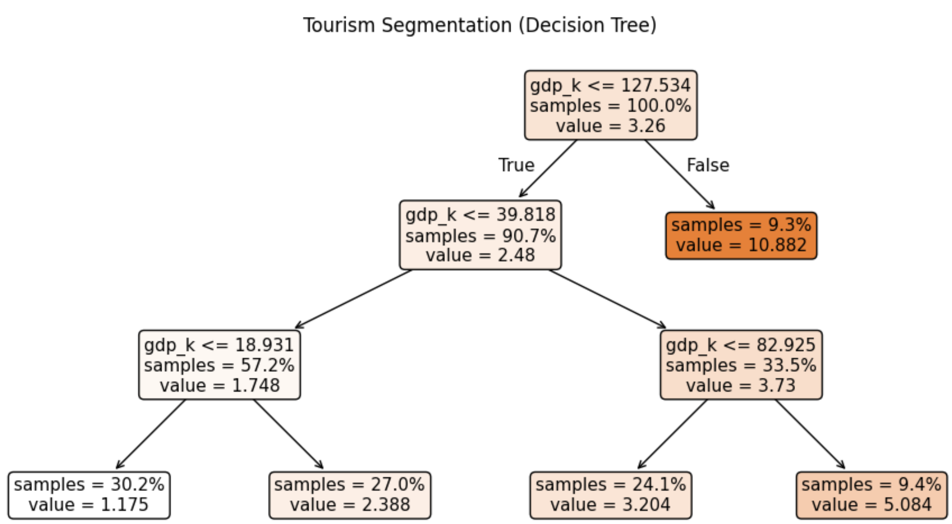

Figure 6: Tourism segmentation using a decision tree

Figure 7 shows the partial dependence plots derived from the random forest model, which represent the average marginal effect of each variable on predicted tourism while holding other variables constant. The relationship between temperature and tourism is weakly non-linear but remains predominantly increasing over most of the observed range, with only mild flattening at intermediate levels. This suggests that higher temperatures are generally associated with higher tourism activity, without a clear reversal at high values in the machine learning specification.

Precipitation exhibits a relatively flat and weak effect across most of its range, suggesting that variations in rainfall have limited influence on tourism activity at the annual level. In contrast, GDP displays a strong and monotonic relationship with tourism, with higher income levels consistently associated with higher predicted tourism activity. The steep increase observed at higher GDP levels indicates that economic capacity plays a central role in structuring tourism demand.

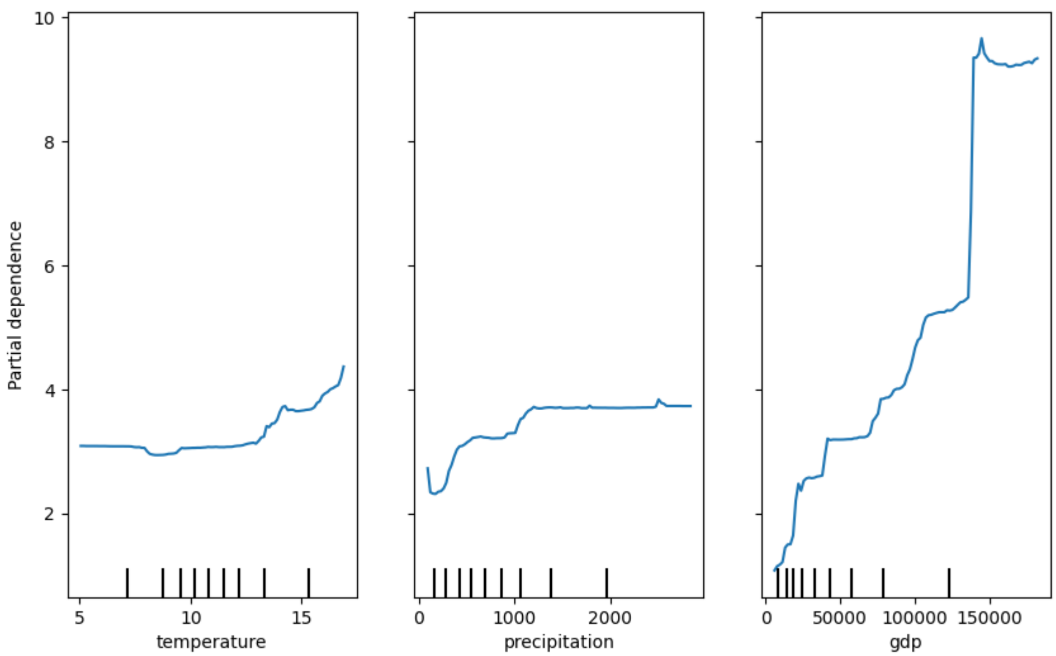

Figure 7: Partial dependence plots for key variables

The random forest model achieves a relatively high out-of-bag performance ($R^2 = 0.659$) with a root mean squared error of 2.44, indicating good predictive capacity. Variable importance measures further confirm the dominance of economic factors, with GDP contributing the most to predictive performance, followed by temperature and precipitation.

These results differ from the econometric findings in an important way. While the fixed-effects models identify a non-linear, inverted U-shaped relationship between temperature anomalies and tourism based on within-region variation, the machine learning results suggest a more gradual and predominantly increasing relationship. One possible explanation is that machine learning models, by construction, are strongly influenced by dominant sources of variation in the data. In this case, cross-sectional differences, particularly in GDP, play a central role and may mask more subtle non-linear effects linked to short-term climate fluctuations.

Overall, these findings highlight that different modelling approaches capture distinct aspects of the data. Econometric models are better suited to isolating within-region dynamics, while machine learning models primarily reflect broader cross-sectional predictive patterns. Taken together, the results reinforce the conclusion that while climate variables exhibit meaningful patterns, they remain secondary relative to structural economic determinants in explaining tourism activity.

10.2 Spatial Network Analysis

To complement the econometric and machine learning analyses, an exploratory spatial network approach is used to investigate the territorial organisation of European regions. The network analysis does not model tourist mobility flows, transport connections, economic interactions, or origin-destination movements between regions. Instead, it provides a descriptive representation of geographic adjacency between NUTS2 regions.

A spatial adjacency network is constructed using NUTS2 boundaries, where regions are represented as nodes and edges are defined by shared borders between neighbouring regions. Two regions are therefore connected whenever they are geographically contiguous, allowing the network to capture the territorial structure of Europe.

The resulting network contains 273 regions connected by 621 spatial adjacency links based on shared NUTS2 boundaries. The network exhibits a low density (0.0167), indicating that most regions are connected only to a limited number of immediate neighbours. The average degree is 4.55, while the presence of 15 connected components reflects the existence of geographically isolated territories, including islands and peripheral regions.

Table 2: Summary statistics of the spatial network

Metric	Value
Nodes	273
Edges	621
Network density	0.0167
Connected components	15
Average degree	4.55

Several centrality measures are computed to identify structurally important regions within the European spatial system. Figure 8 displays the spatial distribution of betweenness centrality.

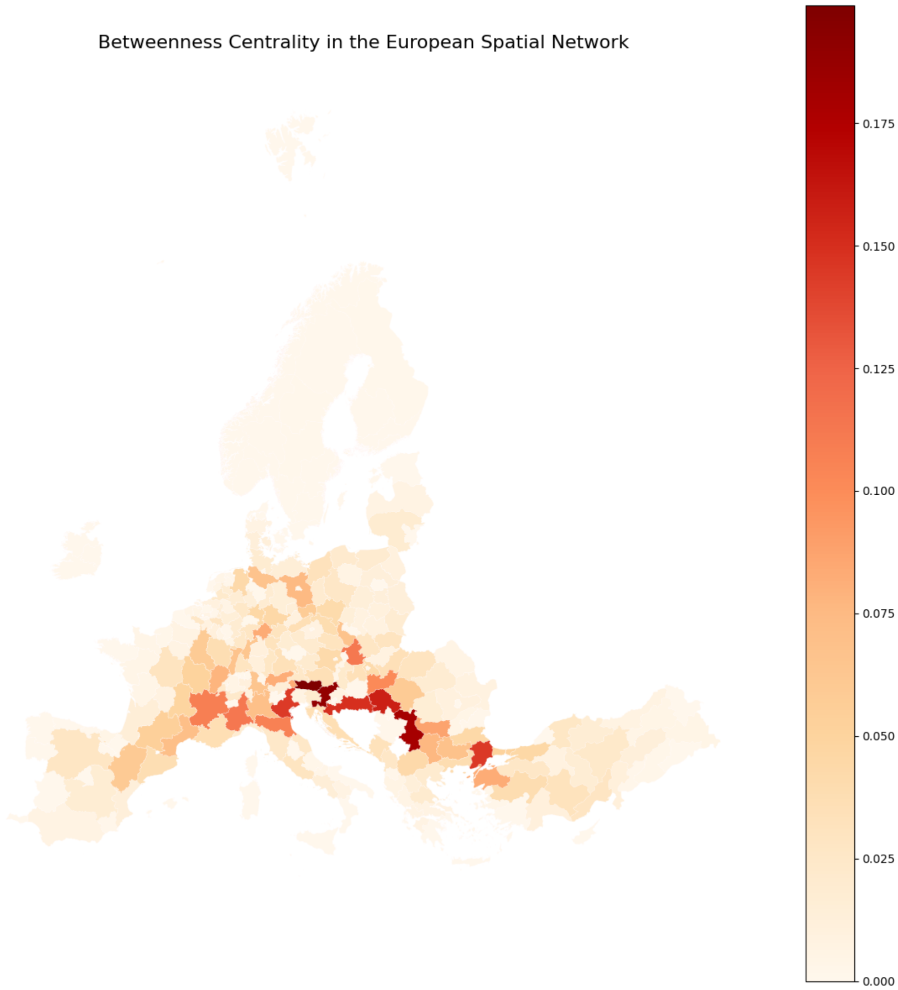

Figure 8: Betweenness centrality in the European spatial network

The highest betweenness values are concentrated in Central and South-Eastern Europe, particularly in Austria, Slovenia, Croatia, Serbia, Northern Italy, and parts of Slovakia. These regions occupy strategic bridging positions connecting major portions of the European spatial network. Their importance arises primarily from geographic location rather than tourism performance.

Table 3 reports the ten regions with the highest betweenness centrality.

Table 3: Top regions by betweenness centrality

Region	Betweenness	Mean log tourism
AT21	0.199	14.919
SI03	0.189	14.098
RS22	0.180	13.301
RS12	0.158	12.903
HR02	0.150	12.019
TR21	0.145	13.670
ITH3	0.144	16.240
ITC1	0.113	15.278
SK02	0.112	13.875
FRK2	0.108	16.417

Community detection using the Louvain algorithm reveals a clear territorial structure. Figure 9 shows that identified communities largely correspond to geographically coherent macro-regions, including Northern Europe, Western Europe, Central Europe, the Balkans, Italy, and Eastern Europe.

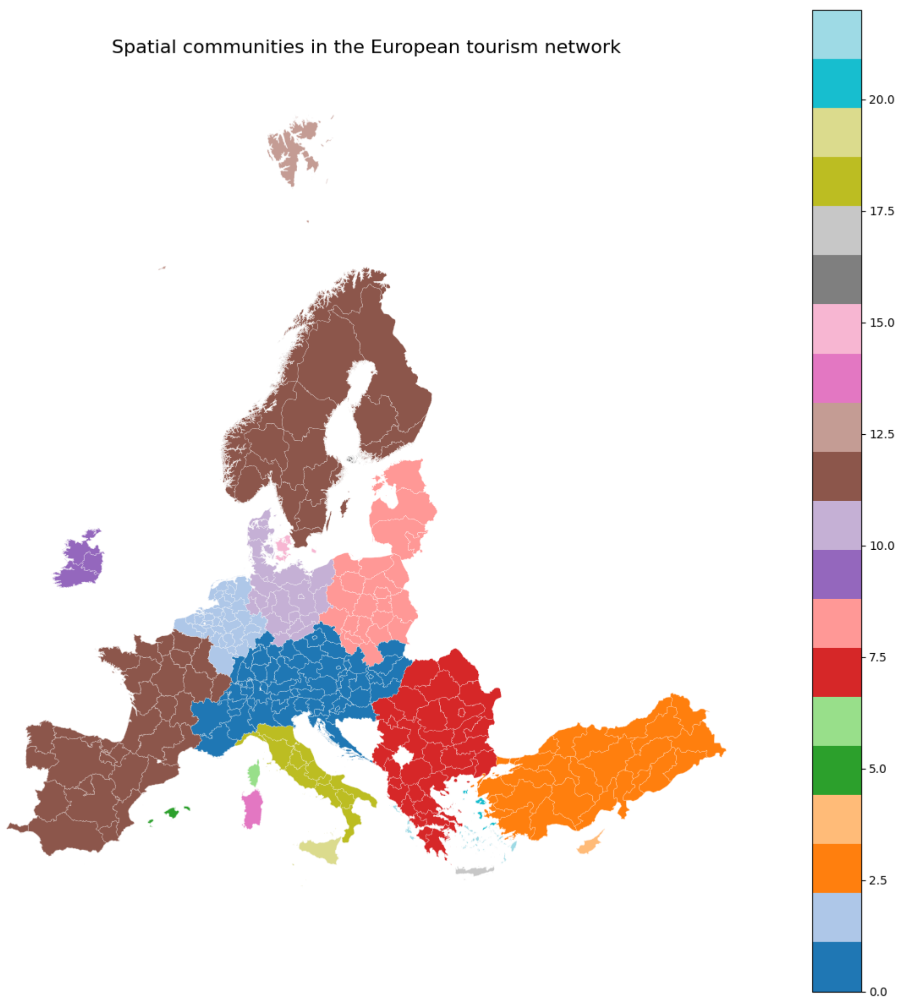

Figure 9: Spatial communities detected using the Louvain algorithm

The emergence of geographically coherent communities suggests that European tourism systems are embedded within broader territorial structures and regional neighbourhood effects. These clusters reflect spatial organisation rather than economic or climatic similarity alone.

Finally, Figure 10 explores the relationship between tourism intensity and network position using betweenness centrality.

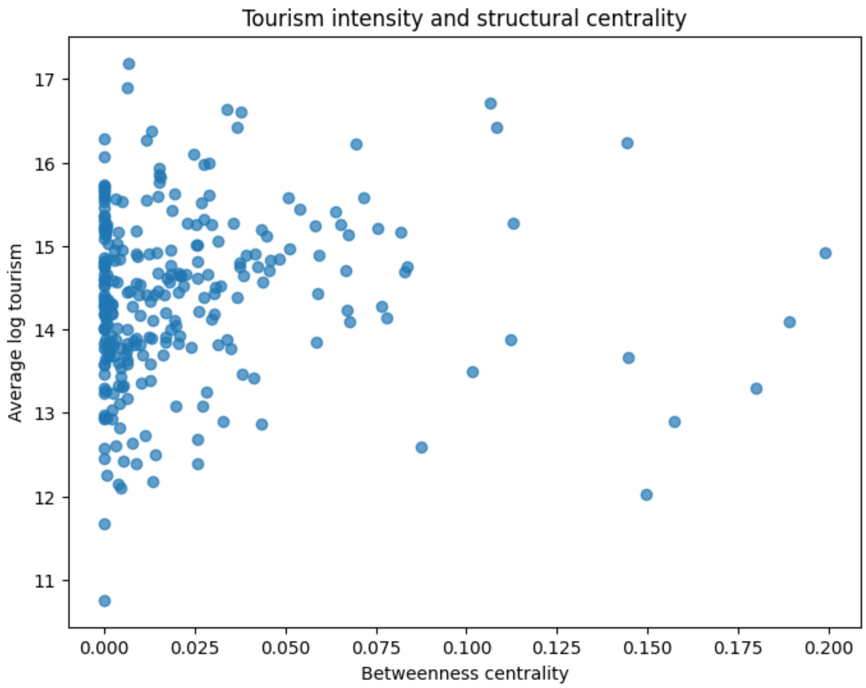

Figure 10: Tourism intensity and structural centrality

The relationship appears positive but weak. Correlation coefficients between tourism intensity and degree, closeness, and betweenness centrality range from 0.088 to 0.105. These values indicate that structurally central regions tend to exhibit slightly higher tourism activity, but network position alone explains only a limited share of variation in tourism outcomes.

Overall, the network analysis highlights the importance of spatial organisation and regional interdependencies in Europe. However, since the network is based only on geographic adjacency, its results should be interpreted as descriptive evidence on territorial structure rather than as an explanation of tourism flows, economic interactions, or regional tourism performance.

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